

GRACE W. TEO

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PROFESSIONAL SUMMARY

Senior UX designer and researcher with more than 15 years of experience leading end-to-end mixed-methods research that informs product strategy, user experience, and organizational decision-making in complex technology environments. Skilled at translating ambiguous customer and business needs into actionable insights that drive product direction, feature prioritization, and measurable outcomes. Experienced partnering with product managers, designers, and engineers in Agile environments.

EXPERIENCE

Quantum Improvements Consulting, Orlando, FL

Lead Scientist (Senior Human Factors & UX Designer/Researcher): Apr 2020 – Feb 2026

- Led end-to-end UX strategy and mixed-methods research across multiple concurrent product programs, partnering with product managers, designers, engineers, and SMEs to define research priorities, reduce uncertainty, and drive roadmap decisions.
- Influenced product strategy by identifying high-value features based on user needs, prioritizing multimodal information presented to users, and collaborating with engineers to define data quality requirements and validation checkpoints that improved product reliability.
- Guided product design of immersive training platforms by identifying instructors' information needs and translating task analysis into learning objectives, behavioral metrics, and AR/VR interaction requirements.
- Directed discovery research that informed features for a collaborative scenario-planning platform supporting distributed decision-making, identifying user needs related to trust, shared understanding, and team coordination.
- Developed and executed evaluation metrics and strategies for AR/VR training solutions, leading pilot studies that demonstrated a 15% improvement in procedural knowledge and 100% positive user engagement compared with traditional training methods, helping secure stakeholder support for broader adoption.
- Accelerated qualitative data processing with AI-assisted workflows (Whisper AI) and identified behavioral measures and patterns most critical for team performance with applied machine learning techniques.
- Mentored 8–10 junior researchers on study design, analysis, and research best practices, strengthening research quality across multiple programs.

Institute for Simulation and Training, University of Central Florida, Orlando, FL

Research Associate (Human Factors): Aug 2013 – Mar 2020

- Led research for the adaption strategy of a human-agent teaming system by modeling cognitive workload and developing of an individualized workload models and decision thresholds identifying high-demand states using physiological measures (e.g., eye-tracking, EEG, ECG, fNIR, TCD).
- Translated findings of Human–Robot Interaction (HRI) research on users' mental models and natural communication patterns into design implications for more intuitive voice- and command-based interfaces.
- Contributed to US. Nuclear Regulatory Commission (NRC)-funded research on workload and error risk in simulation interfaces, identifying tradeoffs between field-of-view fidelity and response time, and input modality effects on selection accuracy to inform interface design decisions.
- Developed reusable research frameworks and standardized methodologies for study execution and analysis, improving consistency and scalability of research across projects.
- Communicated research findings and design recommendations to technical and non-technical stakeholders and mentored junior researchers in study design, data collection, and analysis.

University of Central Florida, Orlando, FL

Human Factors Researcher: Aug 2008 – Jul 2013

- Collaborated on research to inform the design of adaptive driver assistance systems by analyzing driver behavior and cognitive load under varying conditions, generating requirements for system adaptation and feedback mechanisms.
- Conducted research on operator vigilance in high-stakes environments, identifying cognitive and behavioral factors that informed selection, training, and system design for attention and workload management.
- Disseminated research findings through technical reports, conference presentations, and research

DSO National Laboratories, Singapore

Research Scientist (Human Factors): Oct 2005 – Aug 2008

- Led research on decision-making in cyber operations environments, identifying workflow, cognitive, and communication challenges that informed decision-support tools and operational processes.
- Investigated the effects of sleep deprivation and nutraceuticals on human performance, translating findings into recommendations for task design and operator support.

Civil Service College / Public Service Division, Singapore

Senior Psychologist / Psychologist (Industrial-Organizational Psychology): Apr 1996 – Oct 2005

- Evaluated and administered psychometric assessments to support personnel selection and workforce decision-making, ensuring validity and reliability of selection tools used in hiring processes.
- Designed and analyzed organizational surveys supporting leadership development and workforce planning initiatives, translating workforce data into insights for talent and organizational decisions.

EDUCATION

Ph.D., Applied Experimental & Human Factors Psychology: University of Central Florida, 2015

M.A., Applied Experimental & Human Factors Psychology: University of Central Florida, 2013

B.S. (Honors), Psychology: National University of Singapore, 1996

CERTIFICATES & PROFESSIONAL DEVELOPMENT

Data Mining with SAS Certificate: University of Central Florida, 2013

Design for Usability Certificate: University of Central Florida, 2012

IBM AI Product Management Certificate (in progress): Coursera

Google UX Design Certificate (in progress): Coursera

SKILLS

UX Design & Research

Research lifecycle • Data & measurement frameworks • Mixed-methods research • Generative & evaluative research • Quantitative & qualitative synthesis • Psychometrics • Power analysis • Benchmarking • Statistical analysis • Behavioral analytics • A/B testing • Experiments • Interviews • Surveys • Journey mapping • Personas • Task analysis • Think-aloud protocols • Cognitive walk throughs • User flows & interaction design • Information architecture • Design requirements • Prototyping • Usability testing

Strategy & Collaboration

Product strategy • Systems thinking • Human-AI collaboration • Agile product development • Cross-functional collaboration • Workshop facilitation • Executive communication

Analytics & Tools

Python • SQL • R • SPSS • Figma • Power BI • Qualtrics • UserTesting • JIRA • MS Office • AI-based tools (e.g., Whisper AI for speech-to-text transcriptions; Ollama + Mistral + Python for an offline AI agent)